

## Markscheme

Mathematics: analysis and approaches

Practice question bank

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**50. (a)**

Consider the equation  $4^x - 3 \times 2^{x+1} - 16 = 0$ .

Solve the equation for  $x$ .

$$\text{let } u = 2^x, \text{ noting } 4^x = u^2 \text{ and } 3 \times 2^{x+1} = 6u: u^2 - 6u - 16 = 0 \quad (M1) \text{ A1}$$

$$(u - 8)(u + 2) = 0 \Rightarrow u = 8 \text{ or } u = -2 \quad (M1)$$

$$\text{reject } u = -2 \text{ since } 2^x > 0; u = 8 \quad \text{R1}$$

$$2^x = 8 \Rightarrow x = 3 \quad \text{A1}$$

**[5 marks]**

**(b)**

Hence find the value of  $8^x$ .

$$8^3 \quad (M1)$$

$$= 512 \quad \text{A1}$$

**[2 marks]**